

MTH 446: Lecture # 18

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Lecture Span

1. Hyperbolic complex functions

Complex trig

Note that $\sinh$ and $\cosh$ are periodic functions with period $2\pi i$. We can then define

1. $\tanh = \sinh / \cosh$
2. $\coth = \cosh / \sinh$.
3. $\operatorname{sech} = 1 / \cosh$
4. $\operatorname{csch} = 1 / \sinh$

Here, such functions are defined for z values such that the denominator is not zero. We can also define derivatives of the hyperbolic trigs.

Inverse trig functions

$$w = \arcsin(z) \iff z = \sin(w) \quad (1)$$

Note we can rewrite

$$z = \frac{\exp(iw) + \exp(-iw)}{2i} \quad (2)$$

Then, we can solve for w in terms of z , as (letting $t = \exp(iw)$)

$$t^2 - 2izt - 1 = 0 \quad (3)$$

Then,

$$t = \frac{2iz \pm [(2iz)^2 + 4]^{1/2}}{2} \quad (4)$$

Then, you obtain

$$w = \arcsin(z) = -i \log \left(iz + (i - z^2)^{1/2} \right) \quad (5)$$

Defining the domain of definition is difficult. We will just assume that z_0 is defined for $\arcsin$.

Example

Calculate $\arcsin(i)$. Then, using the formula, we obtain

$$\arcsin = -i \log(-1 \pm \sqrt{2}) \quad (6)$$

Then,

$$\log(-1 \pm \sqrt{2}) = \begin{cases} \ln(-1 + \sqrt{2}) + 2\pi ni \\ \ln(1 + \sqrt{2}) + (\pi + 2\pi n)i \end{cases} \quad (7)$$

Note, that we can rewrite this guy as

$$\arcsin(i) = \pi n + (-1)^n i \ln(\sqrt{2} + 1) \quad (8)$$

Derivative of $\arcsin$

We use the single-valued branch of $\log$. Then

$$\arcsin' = \frac{1}{(1 - z^2)^{1/2}} \quad (9)$$

Complex-valued functions of real variable

An interval $I \subseteq \mathbb{R}$ is non-degenerate if it contains two or more points. Therefore, I contains infinitely many points. Consider functions w on I , a non-degenerate interval. Then,

$$w(t) = u(t) + iv(t) \quad (10)$$

For derivatives, we have

$$w' = u' + iv' \quad (11)$$

Also,

1. $D_t(z_0 w) = z_0 w'$
2. $D_t(\exp(z_0 t)) = z_0 \exp(z_0 t)$